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B.Tech. Degree VI Semester Special Supplementary Examination January 2017

ME 1602 MACHINE DESIGN I (2012 Scheme)

Time : 3 Hours

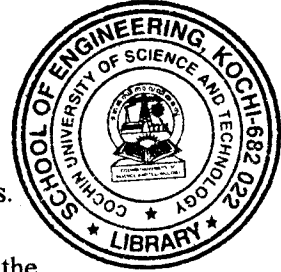
Maximum Marks : 100

(Use of approved Design Data handbook permitted)

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Briefly explain the methods of reducing stress concentration.
- (b) Explain the role of ergonomics in machine design.
- (c) What are the various types of permanent and detachable fastenings?
- (d) How are the key's classified? Also state their applications.
- (e) Explain the modes of failure of a riveted joint.
- (f) Explain A.M. Wahl's factor and its relevance in the design of helical springs.
- (g) Derive an expression for maximum shear stress in parallel fillet welds.
- (h) What are the merits to prefer hollow shafts over solid shaft? Also mention the uses of hollow shaft.



PART B

(4 × 15 = 60)

- II. (a) State and explain any two theories of failure. (5)
- (b) In an elastic material, principal stresses are tensile and compressive and ratio being 4:1. Determine the limiting stress according to different theories of failure, if tension test gives elastic limit of the material as 400 N/mm². Assume Poisson's ratio as 0.3. (10)

OR

- III. (a) Briefly explain the effect of thermal stress in design. (5)
- (b) A steel rod is subjected to a reversed axial load of 180 kN. Find the diameter of the rod for a factor of safety of 2. Neglect column action. The material has an ultimate tensile strength of 1700 N/mm² and yield strength of 910 N/mm². The endurance limit in reversed bending may be assumed to be one half of the ultimate tensile strength. Other correction factors for axial loading = 0.7, for machined surface = 0.8, for size = 0.85, for stress concentration = 1.0. (10)

- IV. (a) What is collar friction in power screws? (5)
- (b) A double threaded power screw, used for lifting the load, has a nominal diameter of 30 mm and a pitch of 6 mm. The coefficient of friction at the screw threads is 0.1. Neglecting collar friction, calculate efficiency of the screw with square threads and Acme threads. (10)

OR

- V. A 15 kW, 960 rpm motor has a M.S. shaft of 40 mm diameter and the extension being 75 mm. The permissible shear and crushing stresses for the M.S. key are 56 MPa and 112 MPa. Design the key in the motor shaft extension. Check the shear strength of the key against the normal strength of the shaft. (15)

(P.T.O.)

- VI. Design a double riveted butt joint with two cover plates for the longitudinal seam of boiler shell 1.5 m in diameter subjected to a steam pressure of 0.95 N/mm^2 . Assume joint efficiency as 75%, allowable tensile stress in the plate 90 MPa, compressive stress 140 MPa and shear stress in the rivet 56 MPa. (15)

OR

- VII. (a) Briefly explain nipping in a leaf spring. (5)
(b) Design a helical compression spring for a maximum load of 1000 N for a deflection of 25 mm using the value of spring index as 5. The maximum permissible shear stress for spring wire is 420 MPa and modulus of rigidity is 84 kN/mm^2 . (10)

- VIII. (a) Write a note on welds subjected to fluctuating loads. (5)
(b) The plates of 10 mm thickness each are to be joined by means of a single riveted double strap butt joint. Determine the rivet diameter, pitch, strap thickness and efficiency of the joint. Take the working stress in tension and compression as 80 MPa and 60 MPa respectively. (10)

OR

- IX. (a) Define equivalent twisting moment and equivalent bending moment with respect to shafts. (5)
(b) A circular shaft is subjected to a bending moment of 3000 Nm and a torque of 10000 Nm. The shaft is made of 45C8 steel having ultimate tensile stress of 700 MPa and ultimate shear stress of 500 MPa. Determine the diameter of the shaft assuming a factor of safety as 6. (10)